2026-07-27--jam99--ThetaI-rewrite-v4.nb

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Confirm that Lekkala2013 Equation 17, rewritten, agrees with the equation I have coded in Python.

From 2023-09-23--jam99--ThetaI-rewrite-v2.nb

We wish to rewrite Equation 17 in Lekkala2013, *J. Chem. Phys.* (2013) 139:184702.
Write out the numerator and denominator in Lekkala2013 θ_l separately, with $\theta 1 = k\, h_s$ and $\theta 2 = \eta\, h_s$.
Lekkala writes

$$\theta_l(k,\, \omega) \,=\, \frac{\epsilon_s(\omega)}{\epsilon_{\text{eff}}(\omega)}\left[-\,\lambda\,\text{Coth}[\theta 1] \,+\, \frac{\text{numerator}}{\text{denominator}}\right]$$

Below we rewrite the part in braces. In my code, I compute

$$\frac{\theta_l(k,\omega)}{\epsilon_s(\omega)} \,=\, \frac{1}{\epsilon_{\text{eff}}(\omega)}\left[-\,\lambda\,\text{Coth}[\theta 1] \,+\, \frac{\text{numerator}}{\text{denominator}}\right]$$

which we recognize as the part in braces divided by $\epsilon_{\text{eff}}(\omega)$.

In[2]:=
```
numerator =
 Sinh[θ1] Sinh[θ2] + α Cosh[θ1] Sinh[θ2] - λ Cosh[θ1] Cosh[θ2] + 2 λ - λ^2 Sinh[θ1]/Sinh[θ2]
```

Out[2]=
```
2 λ - λ Cosh[θ1] Cosh[θ2] - λ^2 Csch[θ2] Sinh[θ1] + α Cosh[θ1] Sinh[θ2] + Sinh[θ1] Sinh[θ2]
```

In[3]:=
```
denominator = Cosh[θ1] Sinh[θ2] + α Sinh[θ1] Sinh[θ2] - λ Sinh[θ1] Cosh[θ2]
```

Out[3]=
```
-λ Cosh[θ2] Sinh[θ1] + Cosh[θ1] Sinh[θ2] + α Sinh[θ1] Sinh[θ2]
```

Divide the denominator by Cosh[θ] to see if that tames the $\theta 1$ divergence.

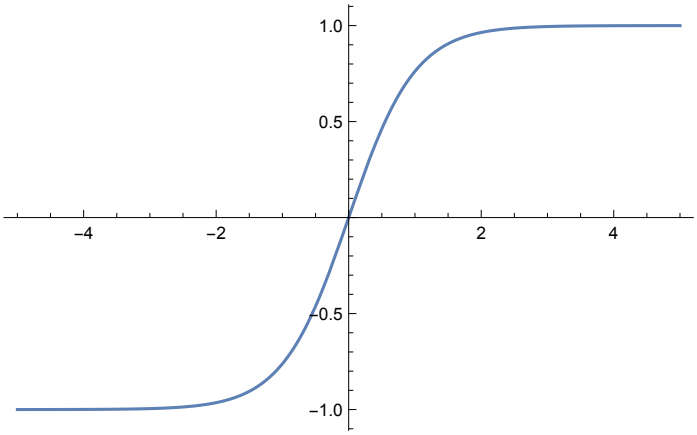
In[4]:=
```
denominator/Cosh[θ1] // Simplify
```

Out[4]=
```
Sinh[θ2] - λ Cosh[θ2] Tanh[θ1] + α Sinh[θ2] Tanh[θ1]
```

It does ...

In[5]:=
```
Plot[{Tanh[θ1]}, {θ1, -5, 5}]
```

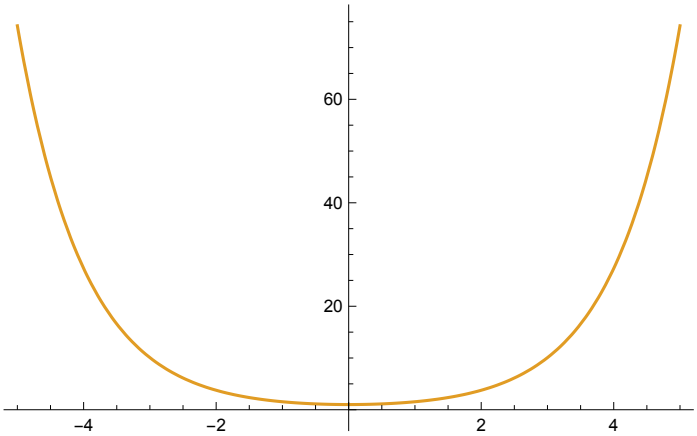
Out[5]=



... but the $\theta 2$ functions in the denominator still diverge.

In[6]:=
```
Plot[{Sinh[θ], Cosh[θ2]}, {θ2, -5, 5}]
```

Out[6]=



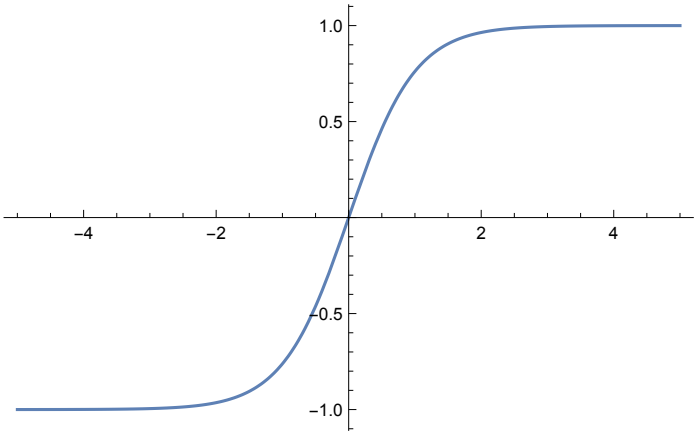
Try dividing the denominator by Cosh[$\theta 1$]Cosh[$\theta 2$].

In[7]:=
```
denominator / (Cosh[θ1] Cosh[θ2]) // Simplify
```

Out[7]=
```
Tanh[Θ2] + Tanh[Θ1] (-λ + α Tanh[Θ2])
```

In[8]:=
```
Plot[Tanh[θ2], {θ2, -5, 5}]
```

Out[8]=



Now, by inspection, both the $\theta 1$ and $\theta 2$ functions in the denominator are well-behaved. What about the numerator?

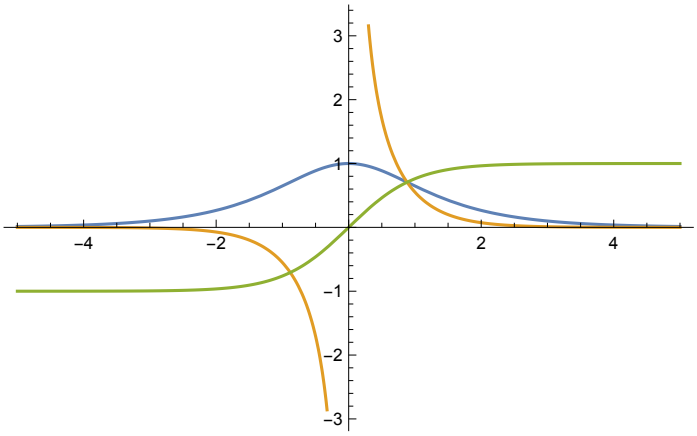
In[9]:=
```
numerator / (Cosh[θ1] Cosh[θ2]) // Simplify
```

Out[9]=
$$-\lambda + 2\,\lambda\,\text{Sech}[\theta 1]\,\text{Sech}[\theta 2] - \lambda^2\,\text{Csch}[\theta 2]\,\text{Sech}[\theta 2]\,\text{Tanh}[\theta 1] + \alpha\,\text{Tanh}[\theta 2] + \text{Tanh}[\theta 1]\,\text{Tanh}[\theta 2]$$

All three $\theta 2$ functions in the numerator are ok at large $\theta 2$. But one of them diverges at small $\theta 2$, which is a problem.

In[10]:=
```
Plot[{Sech[θ2], Csch[θ2] Sech[θ2], Tanh[θ2]}, {θ2, -5, 5}]
```

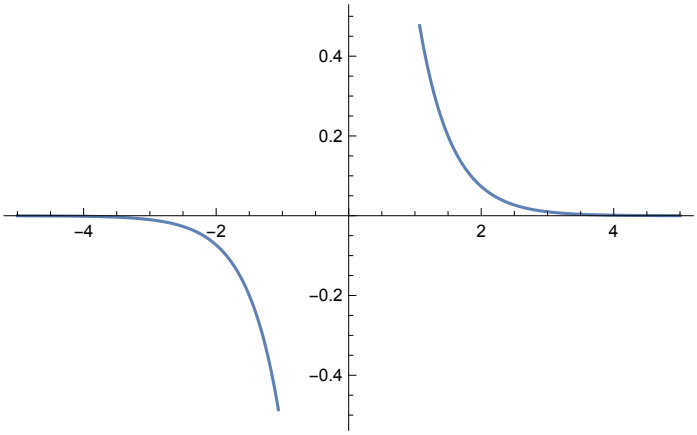
Out[10]=



This function is the culprit.

In[11]:=
```
Plot[Csch[θ2] Sech[θ2], {θ2, -5, 5}]
```

Out[11]=



The argument $\theta 2 = \eta\, h_s$ in the paper, and η does not go to zero. So, in practice, I think the function Csch[$\theta 2$] Sech[$\theta 2$] should be ok.

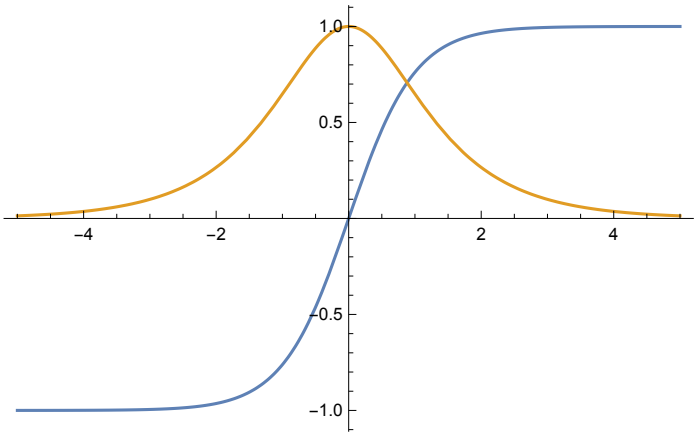
In summary, write the big fraction θ_l as follows

In[12]:=
```
(-λ + 2 λ Sech[θ1] Sech[θ2] - λ^2 Csch[θ2] Sech[θ2] Tanh[θ1] + α Tanh[θ2] + Tanh[θ1] Tanh[θ2])
/ (Tanh[θ2] + Tanh[θ1] (-λ + α Tanh[θ2]))
```

```
;
```

In[13]:=
```
Plot[{Tanh[θ2], Sech[θ2]}, {θ2, -5, 5}]
```

Out[13]=



Python does not have Sech and Csch function. However, note that

In[14]:=
```
1 / Sech[θ1]
```

Out[14]=
```
Cosh[θ1]
```

In[15]:=
```
1 / Csch[θ2]
```

Out[15]=
```
Sinh[θ2]
```

So let us implement θ_l as (we are leaving out the $\epsilon_s/\epsilon_{\text{eff}}$ prefactor here)

```
In[16]:= θI = -λ Coth[θ2] + (Tanh[θ1] Tanh[θ2] + α Tanh[θ2] - λ + 2 λ / (Cosh[θ1] Cosh[θ2]) -
             λ^2 Tanh[θ1] / (Cosh[θ2] Sinh[θ2])) / (Tanh[θ2] + Tanh[θ1] (-λ + α Tanh[θ2])) ;
```

Expand the denominator

```
In[17]:= Tanh[θ2] + Tanh[θ1] (-λ + α Tanh[θ2]) // Expand
Out[17]= -λ Tanh[θ1] + Tanh[θ2] + α Tanh[θ1] Tanh[θ2]
```

Check that this expression agrees with what we started with.

```
In[18]:= θI - (-λ Coth[θ2] + numerator / denominator) // FullSimplify
Out[18]= 0
```

This form is well behaved because $\theta 2$ does not go to zero. Check the large h_s limit, where $\theta 1$ and $\theta 2$ go to infinity.

```
In[19]:= Limit[θI, {θ1 → Infinity, θ2 → Infinity}]
Out[19]= 1 - λ
```

Check the Mathematica result against my code

Forward substitution

Rewrite the *Mathematica* expression in terms of my variables p1 through p7.

```
In[35]:= θI
Out[35]= -λ Coth[θ2] + (-λ + 2 λ Sech[θ1] Sech[θ2] - λ^2 Csch[θ2] Sech[θ2] Tanh[θ1] +
           α Tanh[θ2] + Tanh[θ1] Tanh[θ2]) / (Tanh[θ2] + Tanh[θ1] (-λ + α Tanh[θ2]))
```

Divide the numerator in the second term by ϵ_{eff}^2 and use the variables defined in my code to rewrite the answer:

```
In[65]:= (1 / ϵ$eff^2 (-λ + 2 λ Sech[θ1] Sech[θ2] -
             λ^2 Csch[θ2] Sech[θ2] Tanh[θ1] + α Tanh[θ2] + Tanh[θ1] Tanh[θ2]) // Expand) /.
           {1 / ϵ$eff^2 → p2, α / ϵ$eff^2 → p3, λ / ϵ$eff^2 → - p4, λ^2 / ϵ$eff^2 → -p5}
Out[65]= p4 - 2 p4 Sech[θ1] Sech[θ2] +
          p5 Csch[θ2] Sech[θ2] Tanh[θ1] + p3 Tanh[θ2] + p2 Tanh[θ1] Tanh[θ2]
```

This looks like the numerator in my code.

Divide the denominator in the second term by ϵ_{eff} and use the variables in my code to rewrite the answer:

In[67]:=
$$\left(\frac{\text{Tanh}[\theta 2]+\text{Tanh}[\theta 1]\ (-\lambda+\alpha\ \text{Tanh}[\theta 2])}{\epsilon\$\text{eff}}\ //\ \text{Expand}\right)\ /.\ \left\{-\frac{\lambda}{\epsilon\$\text{eff}}\to \text{p1},\ \frac{\alpha}{\epsilon\$\text{eff}}\to \text{p7},\ \frac{1}{\epsilon\$\text{eff}}\to \text{p6}\right\}\ //\ \text{Simplify}$$

Out[67]=
$$\text{p6 Tanh}[\theta 2]+\text{Tanh}[\theta 1]\ (\text{p1}+\text{p7 Tanh}[\theta 2])$$

This looks like the denominator in my code.

Backward substitution

Now write out my equation with $\theta 1$ and $\theta 2$ swapped. My coded equation θI\$code is really θ_I/ϵ_s, as note above.

Substitute for the variables p1 through p7.

In[78]:=
```
θI$code =
  p1 Coth[θ2] + (p2 Tanh[θ1] Tanh[θ2] + p3 Tanh[θ2] + p4 - 2 p4 Sech[θ1] Sech[θ2] +
      p5 Tanh[θ1] Sech[θ2] Csch[θ2]) / (p6 Tanh[θ2] + Tanh[θ1] (p1 + p7 Tanh[θ2]));
```

In[79]:=
$$\theta\text{I\$code\$sub}=\theta\text{I\$code}\ /.\ \left\{\text{p1}\to -\frac{\lambda}{\epsilon\$\text{eff}},\ \text{p2}\to \frac{1}{\epsilon\$\text{eff}^2},\ \text{p3}\to \frac{\alpha}{\epsilon\$\text{eff}^2},\ \text{p4}\to -\frac{\lambda}{\epsilon\$\text{eff}^2},\ \text{p5}\to -\frac{\lambda^2}{\epsilon\$\text{eff}^2},\ \text{p6}\to \frac{1}{\epsilon\$\text{eff}},\ \text{p7}\to \frac{\alpha}{\epsilon\$\text{eff}}\right\}$$

Out[79]=
$$-\frac{\lambda\ \text{Coth}[\theta 2]}{\epsilon\$\text{eff}}+\frac{-\frac{\lambda}{\epsilon\$\text{eff}^2}+\frac{2\lambda\ \text{Sech}[\theta 1]\ \text{Sech}[\theta 2]}{\epsilon\$\text{eff}^2}-\frac{\lambda^2\ \text{Csch}[\theta 2]\ \text{Sech}[\theta 2]\ \text{Tanh}[\theta 1]}{\epsilon\$\text{eff}^2}+\frac{\alpha\ \text{Tanh}[\theta 2]}{\epsilon\$\text{eff}^2}+\frac{\text{Tanh}[\theta 1]\ \text{Tanh}[\theta 2]}{\epsilon\$\text{eff}^2}}{\frac{\text{Tanh}[\theta 2]}{\epsilon\$\text{eff}}+\text{Tanh}[\theta 1]\left(-\frac{\lambda}{\epsilon\$\text{eff}}+\frac{\alpha\ \text{Tanh}[\theta 2]}{\epsilon\$\text{eff}}\right)}$$

Check that the resulting expression agrees with the *Mathematica* result. As noted above, we should divide θI, "the part in braces", by ϵ_{eff} to obtain θ_I/ϵ_s.

In[81]:=
$$\frac{\theta\text{I}}{\epsilon\$\text{eff}}-\theta\text{I\$code\$sub}\ //\ \text{Simplify}$$

Out[81]=
$$0$$